MIL-M-38510/23A
14 July 1972
SUPERSEDING
MIL-M-38510/23(USAF)
30 September 1971

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL,

TTL, HIGH SPEED, NAND GATES

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic, silicon, TTL, high speed, positive NAND logic gating microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/023	01	B	A	C
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	Single, 8-input positive NAND gate
02	Dual, 4-input positive NAND gate
03	Triple, 3-input positive NAND gate
04	Quadruple, 2-input positive NAND gate
05	Hex, 1-input inverter gate
06	Quadruple, 2-input positive NAND gate (open collector output)
07	Dual, 4-input positive NAND gate (open collector output)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Case outline A - F-1, (14-pin, 1/4" x 1/4" flat pack)
Case outline B - F-3, (14-pin, 1/4" x 1/8" flat pack)
Case outline C - D-1, (14-pin, 1/4" x 3/4" dual-in-line pack)
Case outline D - F-2, (14-pin, 1/4" x 3/8" flat pack)

1.2.4 Absolute maximum ratings.

Supply voltage range	-0.5 Vdc to 7.0 Vdc
Input voltage range	-1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Maximum power dissipation per gate, P_D [1/]	198 mWdc
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case	θ_{JC} = {0.09°C/mW for flat pack; 0.08°C/mW for dual-in-line pack}
Junction temperature	T_J = 175°C

1/ Must withstand the added P_D due to short circuit conditions (e.g. IOS) at one output for 5 seconds duration.

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1.2.5 Recommended operating conditions.

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc
Normalized fanout (each output) 1/	10 maximum
Ambient operating temperature range	-55°C to 125°C

2. APPLICABLE DOCUMENT

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. In the event of conflict between MIL-M-38510 and this detail specification, this detail specification shall govern.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3, or equivalent.

3.2.4 Case outline.

3.2.4.1 Case outline A. Case outline A shall be in accordance with F-1 of Appendix C to MIL-M-38510.

3.2.4.2 Case outline B. Case outline B shall be in accordance with F-3 of Appendix C to MIL-M-38510.

3.2.4.3 Case outline C. Case outline C shall be in accordance with D-1 of Appendix C to MIL-M-38510.

3.2.4.4 Case outline D. Case outline D shall be in accordance with F-2 of Appendix C to MIL-M-38510.

1/ Device will fanout to both high and low states to the specified number of inputs of the same device type as that being tested.

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3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits		
				Min	Max	Units
High-level output voltage	V_{OH}	V_{CC} = 4.5V, V_{IN} = 0.8V I_{OH} = -500 µA 1/	01, 02, 03, 04, 05	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5V, I_{OL} = 20 mA V_{IN} = 2.0V for all inputs of gate under test 1/	All		0.4	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5V, I_{IN} = -12 mA T_A = 25°C	All		-1.5	Volts
Maximum collector cut-off current	I_{CEX}	V_{CC} = 4.5V, V_{IN} = 0.8V V_{OH} = 5.5V	06, 07		250	µA
High-level input current	I_{IH1}	V_{CC} = 5.5V, V_{IN} = 2.4V 2/	All		50	µA
High-level input current	I_{IH2}	V_{CC} = 5.5V, V_{IN} = 5.5V 2/	All		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5V, V_{IN} = 0.4V 1/	All	-1.0	-2.0	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5V 2/ 3/	01, 02, 03, 04, 05	-40	-100	mA
High-level supply current per gate	I_{CCH}	V_{CC} = 5.5V 2/ V_{IN} = 0V	All		4.2	mA
Low-level supply current per gate	I_{CCL}	V_{CC} = 5.5V 1/ V_{IN} = 5.5V	All		10	mA
Propagation delay time, high-to-low-level	t_{PHL}	C_L = 50 pF, R_L = 280Ω	01	2	18	ns
			02, 03, 04, 05	2	16	ns
			06, 07	3	18	ns
Propagation delay time, low-to-high-level	t_{PLH}	C_L = 50 pF, R_L = 280Ω	01	2	16	ns
			02, 03, 04, 05	2	15	ns
			06, 07	3	21	ns

1/ All unspecified inputs at 5.5 volts.
2/ All unspecified inputs grounded.
3/ Not more than one output should be shorted at a time.

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3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

TABLE II. Electrical test requirements.

MIL-STD-883 Test requirements	Subgroups (see table III)		
	Class A devices	Class B devices	Class C devices
Interim electrical parameters (Pre Burn-In) (Method 5004)	1	1	Not reqd
Final electrical test parameters (Method 5004)	1*, 2, 3, 9	1*, 2, 3, 9	1
Group A test requirements (Method 5005)	1, 2, 3, 9, 10, 11	1, 2, 3, 9	1, 2, 3, 9
Group B and C end point electrical parameters (Method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for Group C periodic inspections	None	10, 11	None

* PDA applies to subgroup 1 (see 4.3 (h)).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.3 herein. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

(a) Country of origin.

4. PRODUCT ASSURANCE PROVISIONS.

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and Method 5005 of MIL-STD-883 except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, and C inspections (see 4.4.1, 4.4.2, and 4.4.3). After qualification of one or more electrically and structurally similar types with a single lead finish, other lead finishes of the same case outline may be qualified by submitting a single type in the qualified case outline to the group B, subgroup 3 test and the group C, subgroup 1, 3, and 4 test.

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4.3 Screening. Screening shall be in accordance with Method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

(a) Test samples for the group B bond strength test specified in Method 5005 of MIL-STD-883 may, at the manufacturer's option, be randomly selected immediately following the internal visual (precap) inspection and prior to sealing (see 4.4.2(b)).
(b) Temperature cycling (Method 1010 of MIL-STD-883).
(1) Omit seal test as post-test measurements.
(c) Thermal shock (Method 1011 of MIL-STD-883), when substituted for temperature cycling.
(1) Omit seal test as post-test measurements.
(d) Burn-in test (Method 1015 of MIL-STD-883).
(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
(2) T_A = 125°C minimum.
(e) Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of Method 5004 of MIL-STD-883 may be omitted.
(f) Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
(g) External visual inspection shall not include measurement of case and lead dimensions.
(h) Percent-defective allowable (PDA) - The PDA is specified as 5 percent for class A devices and 10 percent for class B devices based on failures from Group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of Group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups and LTPD values shown in table I of Method 5005 of MIL-STD-883 and as follows:

(a) Tests shall be as specified in table II.
(b) Subgroups 4, 5, 6, 7, and 8 shall be omitted.

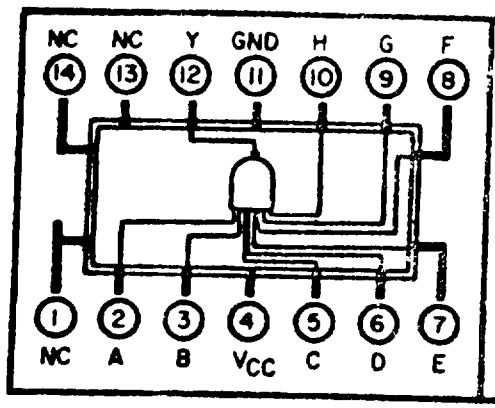
4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table II of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Bond strength test may be conducted on samples collected prior to sealing (see 4.3(a)).

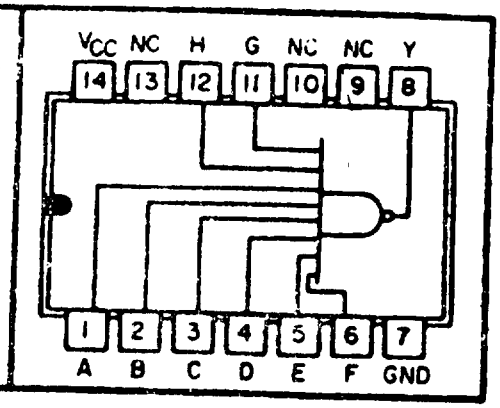
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Device type 01

Case A, B and D

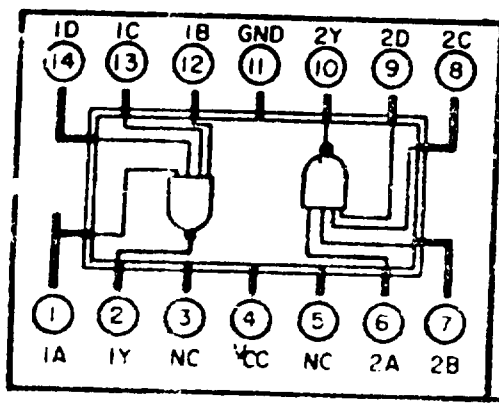


Case C

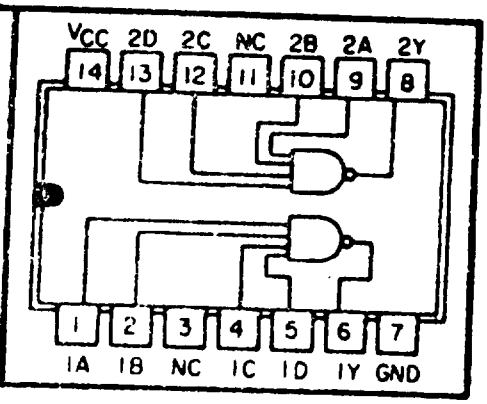


Device type 02 and 07

Case A, B and D

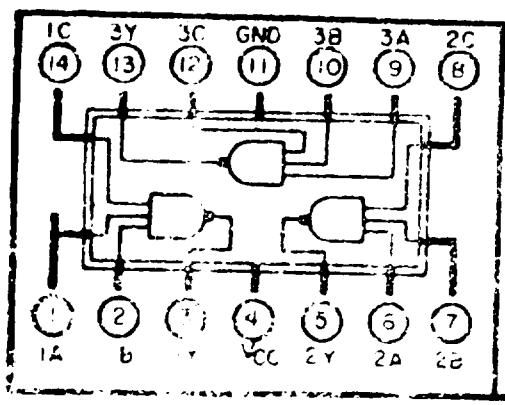


Case C



Device type 03

Case A, B and D



Case C

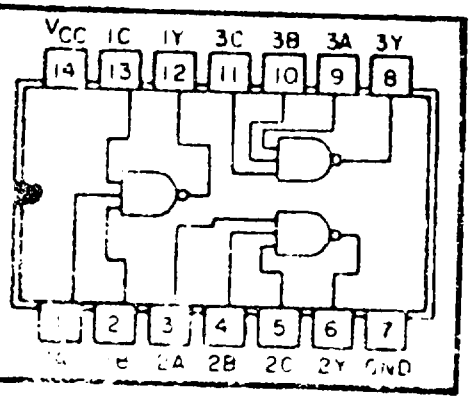
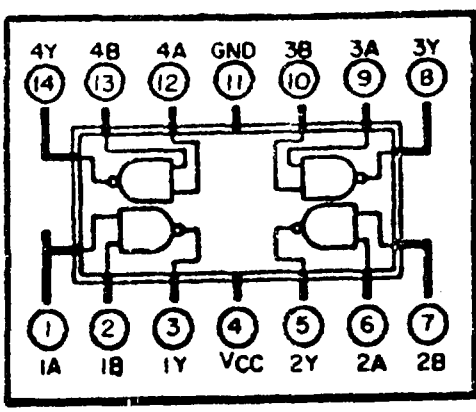


FIGURE 1. Logic diagram and terminal connections (top views).

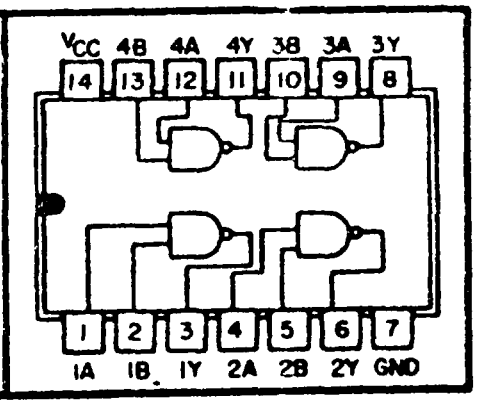
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Device type 04 and 06

Case A, B and D

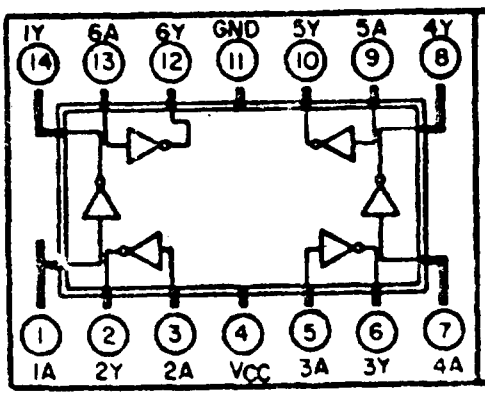


Case C



Device type 05

Case A, B and D



Case C

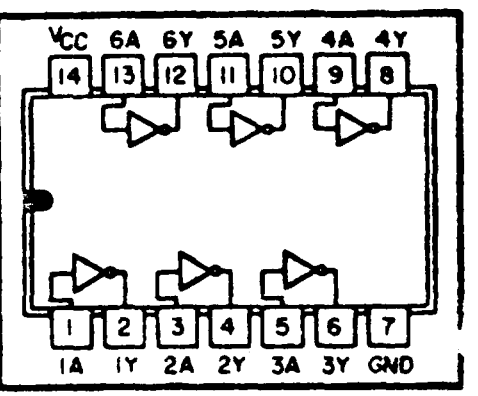


FIGURE 1. Logic diagram and terminal connections (top views) - Continued.

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Device type 01

Truth table								
Input								Output
A	B	C	D	E	F	G	H	Y
H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.								

Positive logic $Y = \overline{ABCDEFGH}$

Device type 05

Truth table each gate	
Input	Output
A	Y
L	H
H	L

Positive logic $Y = \overline{A}$

Device type 03

Truth table			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic $Y = \overline{ABC}$

Device types 02 and 07

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

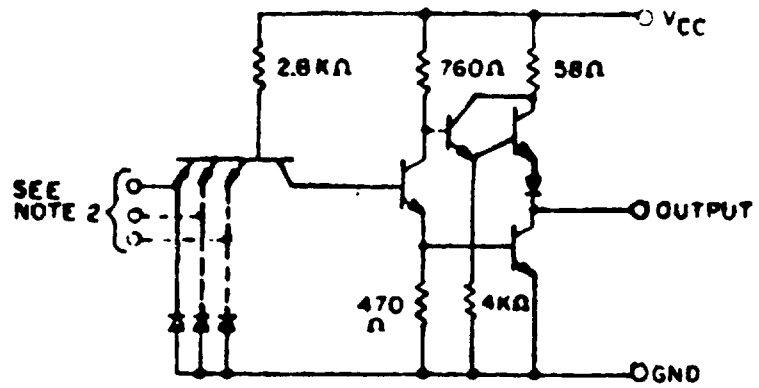
Device types 04 and 06

Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

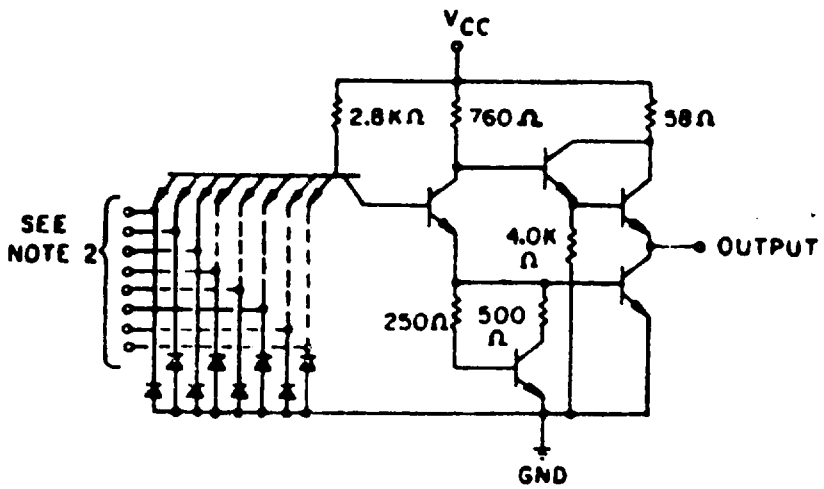
Positive logic $Y = \overline{AB}$

FIGURE 2. Truth tables and logic equations

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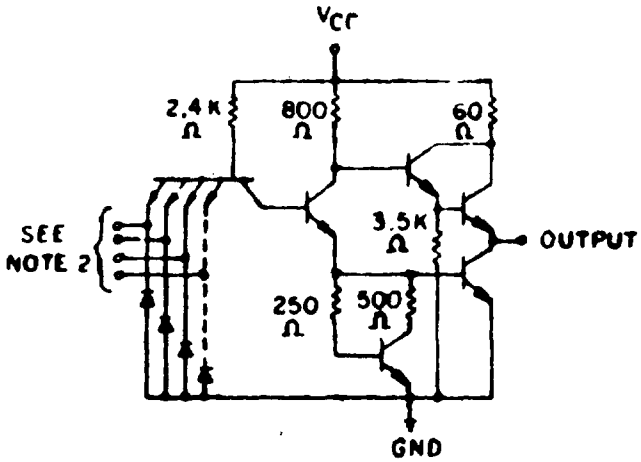
CIRCUIT A



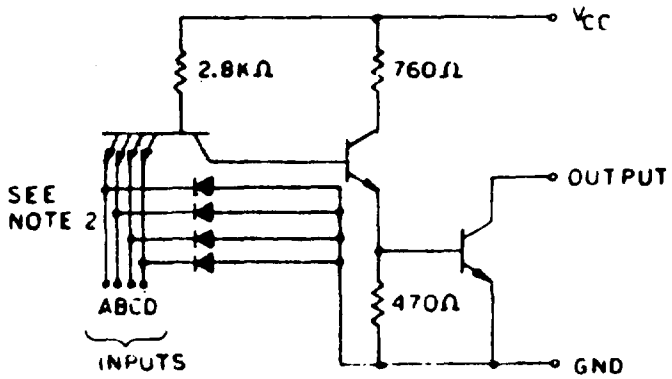
CIRCUIT B

FIGURE 3. Schematic circuit.

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CIRCUIT C

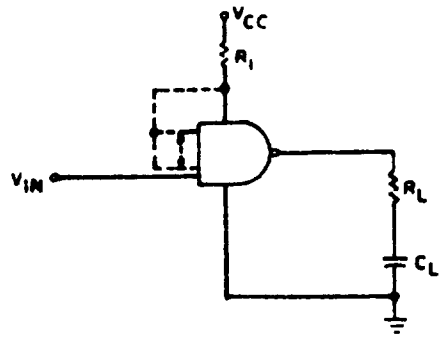


CIRCUIT D

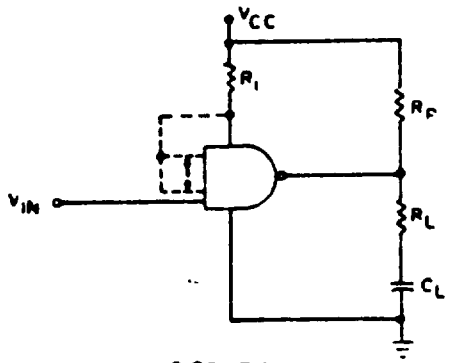
NOTES:
1. Circuits A, B, C, and D are the only acceptable variations in the functional schematic for device types 01, 02, 03, 04, 05, 06, and 07. Circuit A applies to types 01 thru 05. Circuit B applies to types 01 and 05. Circuit C applies to device types 02, 03, and 04, and Circuit D applies to device types 06 and 07.
2. Number of inputs is determined by logic configuration shown in figure 1.
3. All resistance values shown are nominal.

FIGURE 3. Schematic circuits - Continued.

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CIRCUIT A

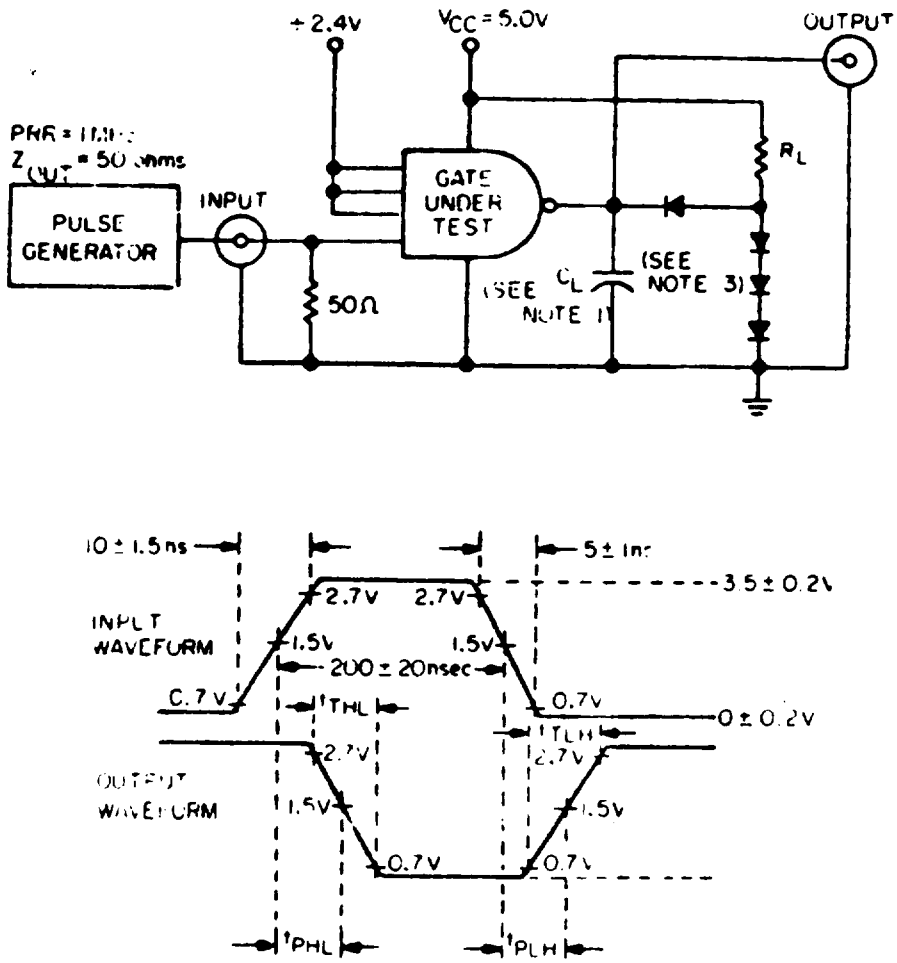


CIRCUIT B

NOTES:
1. One input of each gate shall be driven. The remaining inputs of each gate shall be connected to V_{CC} through the appropriate current limiting resistors.
2. V_{CC} and R_1 are such that the minimum voltage at the device terminals is 5 volts.
3. If parallel excitation is used, the clock frequency shall be 100 KHz with V_{IN} = 3 volts minimum and minimum duty cycle of 50%.
4. R_L = 120 Ω ± 5%; C_L = .047 µf ± 10%; R_P = 300 Ω ± 5%.
5. Circuit A shall be used for device types 01, 02, 03, 04 and 05.
   Circuit B shall be used for device types 06 and 07.

FIGURE 4. Burn-in and life test circuits.

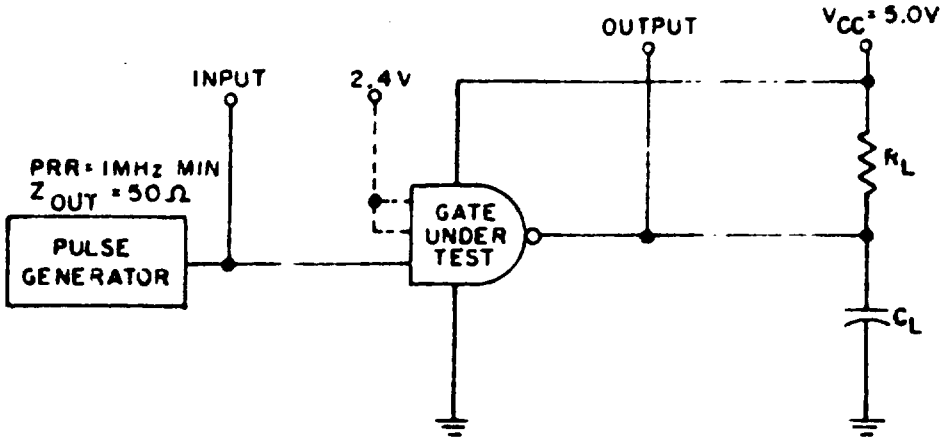
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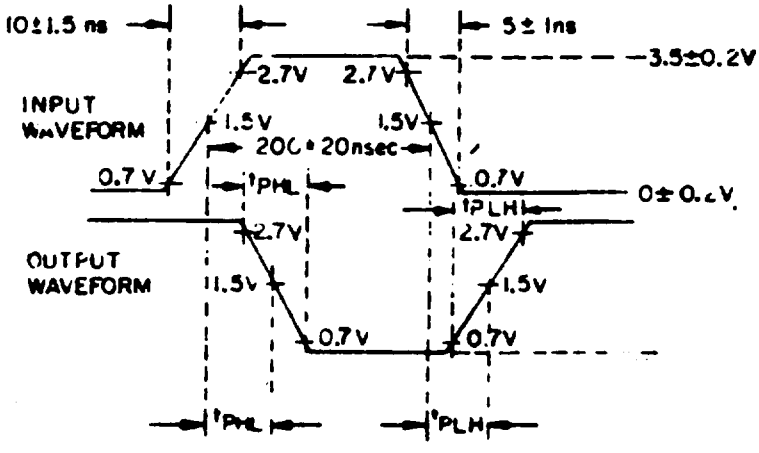


NOTES:
1. C_L = 50 pf minimum, including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 280Ω ± 5%.

FIGURE 5. Switching time test circuit except for open collector circuits.

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NOTES:
1. C_L = 50 pf minimum including scope probe, wiring, and stray capacitance, without package in test fixture.
2. Voltage measurements are to be made with respect to network ground terminal.
3. All diodes are 1N3064 or equivalent.
4. R_L = 280Ω ± 5%.

FIGURE 6. Switching time test circuit for open collector circuits.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case X, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits Min	Max	Unit
			Case C	13	1	2	14	3	4	5	6	11	12	7	8	9	10				
			Test No.	NC	A	B	V_{CC}	C	D	E	Y	G	H	GND	Y	NC	NC				
1	V_{OL}	3007	1		2.0 v	2.0 v	4.5 v	2.0 v	2.0 v	2.0 v	2.0 v	2.0 v	2.0 v	GND	20 mA			Y		0.4	V
T_A = 25°C	V_{OH}	3006	2		0.8 v	0.8 v		0.8 v	0.8 v	0.8 v	0.8 v	0.8 v	0.8 v		-.8 mA			Y	2.4		
	V_{OH}		3		5.5 v	0.8 v		5.5 v	5.5 v												
	V_{OH}		4			5.5 v		0.8 v													
	V_{OH}		5					5.5 v	0.8 v												
	V_{OH}		6						5.5 v	0.8 v											
	V_{OH}		7							5.5 v	0.8 v										
	V_{OH}		8								5.5 v	0.8 v									
	V_{OH}		9								5.5 v	5.5 v	0.8 v								
	V_{CX}	3011	10		GND	GND	5.5 v	GND	GND	GND	GND	GND	GND		GND				-40	-100	mA
	I_{IH1}	3010	11		2.4 v	GND		GND	GND	GND	GND	GND	GND					A		50	μA
	I_{IH1}		12		GND	2.4 v		GND										B			
	I_{IH1}		13			GND		2.4 v	2.4 v									C			
	I_{IH1}		14					GND		2.4 v								D			
	I_{IH1}		15						GND		2.4 v							E			
	I_{IH1}		16							GND		2.4 v						F			
	I_{IH1}		17								GND		2.4 v					G			
	I_{IH1}		18									GND						H			
	I_{IH2}		19		5.5 v	5.5 v							GND					A		100	
	I_{IH2}		20		GND	GND		5.5 v										B			
	I_{IH2}		21					GND	5.5 v									C			
	I_{IH2}		22						GND	5.5 v								D			
	I_{IH2}		23							GND	5.5 v							E			
	I_{IH2}		24								GND	5.5 v						F			
	I_{IH2}		25									GND	5.5 v					G			
	I_{IH2}		26										GND					H			
	I_{IL}	3009	27		0.4 v	5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v					A	-1.0	-2.0	mA
	I_{IL}		28		5.5 v	0.4 v		5.5 v	5.5 v									B			
	I_{IL}		29			5.5 v		0.4 v										C			
	I_{IL}		30					5.5 v	0.4 v									D			
	I_{IL}		31						5.5 v	0.4 v								E			
	I_{IL}		32							5.5 v	0.4 v							F			
	I_{IL}		33								5.5 v	0.4 v						G			
	I_{IL}		34									5.5 v	0.4 v					H			
	I_{CCL}	3005	35		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND				V_{CC}		10	mA
	I_{CCH}	3005	36		GND	GND	5.5 v	GND	GND	GND	GND	GND	GND					V_{CC}		4.2	
	V_{IC}		37		-12 mA		4.5 v											A		-1.5	V
			38			-12 mA												B			
			39					-12 mA										C			
			40						-12 mA									D			
			41							-12 mA								E			
			42								-12 mA							F			
			43									-12 mA						G			
			44										-12 mA					H			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
9	t_{PHL}	3003	45		IN	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	OUT			A to Y	2	14	ns
T_A = 25°C	t_{PLH}	(Fig. 5)	46															A to Y	2	12	
10	t_{PHL}	3003	47		IN	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	OUT			A to Y	2	16	
T_A = 125°C	t_{PLH}	(Fig. 5)	48															A to Y	2	16	
11	t_{PHL}	3003	49		IN	2.4 v	5.0 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	2.4 v	GND	OUT			A to Y	2	18	
T_A = -55°C	t_{PLH}	(Fig. 5)	50															A to Y	2	14	

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 6 / 1Y	3 / 3 / NC	4 / 14 / VCC	5 / 11 / NC	6 / 9 / 2A	7 / 10 / 2B	8 / 12 / 2C	9 / 13 / 2D	10 / 8 / 2Y	11 / 7 / GND	12 / 2 / 1B	13 / 4 / 1C	14 / 5 / 1D	Meas. terminal	Test limits Min	Max	Unit
1 TA = 25°C	V_{OL}	3007	1	2.0 v	20 mA		4.5 v		5.5 v	5.5 v	5.5 v	5.5 v		GND	2.0 v	2.0 v	2.0 v	1Y		.4	V
	V_{OL}	↓	2	5.5 v	↓		↓		2.0 v	2.0 v	2.0 v	2.0 v	20 mA	↓	5.5 v	5.5 v	5.5 v	2Y		.4	↓
	V_{OH}	3006	3	0.8 v	-.5 mA				5.5 v	5.5 v	5.5 v	5.5 v			5.5 v			1Y	2.4		
	V_{OH}		4	5.5 v	-.5 mA										0.8 v			1Y			
	V_{OH}		5		-.5 mA										5.5 v			1Y			
	V_{OH}		6		-.5 mA											0.8 v	0.8 v	1Y			
	V_{OH}		7						0.8 v				-.5 mA			5.5 v	5.5 v	2Y			
	V_{OH}		8						5.5 v	0.8 v			-.5 mA					2Y			
	V_{OH}		9							5.5 v	0.8 v		-.5 mA					2Y			
	V_{OH}	↓	10				↓			5.5 v	5.5 v	0.8 v	-.5 mA		↓	↓	↓	2Y	↓		↓
	I_{OS}	3011	11	GND	GND		5.5 v								GND	GND	GND	1Y	-40	-100	mA
	I_{OS}	↓	12	↓	↓		↓		GND	GND	GND	GND	GND		↓	↓	↓	2Y	-40	-100	mA
	I_{IH1}	3010	13	2.4 v											GND	GND	GND	1A		50	µA
	I_{IH1}		14	GND											2.4 v	GND	GND	1B			
	I_{IH1}		15												GND	2.4 v	GND	1C			
	I_{IH1}		16													GND	2.4 v	1D			
	I_{IH1}		17						2.4 v								GND	2A			
	I_{IH1}		18						GND	2.4 v								2B			
	I_{IH1}		19							GND	2.4 v							2C			
	I_{IH1}		20	↓							GND	2.4 v						2D		↓	
	I_{IH2}		21	5.5 v								GND						1A		100	
	I_{IH2}		22	GND											5.5 v			1B			
	I_{IH2}		23												GND	5.5 v		1C			
	I_{IH2}		24													GND	5.5 v	1D			
	I_{IH2}		25						5.5 v								GND	2A			
	I_{IH2}		26						GND	5.5 v								2B			
	I_{IH2}		27							GND	5.5 v							2C			
	I_{IH2}	↓	28	↓			↓				GND	5.5 v			↓	↓	↓	2D		↓	
	I_{IL}	3009	29	0.4 v			5.5 v		5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	5.5 v	5.5 v	1A	-1.0	-2.0	mA
			30	5.5 v											0.4 v	5.5 v		1B			
			31												5.5 v	0.4 v		1C			
			32													5.5 v	0.4 v	1D			
			33						0.4 v								5.5 v	2A			
			34						5.5 v	0.4 v								2B			
			35							5.5 v	0.4 v							2C			
		↓	36	↓			↓			5.5 v	5.5 v	0.4 v			↓	↓	↓	2D	↓	↓	↓
	I_{CCL}	3005	37							5.5 v	5.5 v	5.5 v						VCC		20	mA
↓	I_{CCH}	3005	38	GND			↓		GND	GND	GND	GND		↓	GND	GND	GND	VCC		8.4	mA

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Cases A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 6 / 1Y	3 / 3 / NC	4 / 14 / VCC	5 / 11 / NC	6 / 9 / 2A	7 / 10 / 2B	8 / 12 / 2C	9 / 13 / 2D	10 / 8 / 2Y	11 / 7 / GND	12 / 2 / 1B	13 / 4 / NC	14 / 5 / 1D	Mode, terminal	Test limits: Min	Max	Unit
1 T_A = 25°C	V_{IC}		39 40 41 42 43 44 45 46	-12 mA			4.5 v		-12 mA	-12 mA	-12 mA	-12 mA		GND	-12 mA	-12 mA	-12 mA	1A 1B 1C 1D 2A 2B 2C 2D		-1.5	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
9 T_A = 25°C	t_{PLH} t_{PHL}	3003 (Fig. 5)	47 48	IN	OUT		5.0 v		IN	2.4 v	2.4 v	2.4 v	OUT	GND	2.4 v	2.4 v	2.4 v	1A to 1Y 2A to 2Y	2	12 12	ns
	t_{PLH} t_{PHL}		49 50	IN	OUT				IN	2.4 v	2.4 v	2.4 v	OUT		2.4 v	2.4 v	2.4 v	1A to 1Y 2A to 2Y		13 12	
10 T_A = 125°C	t_{PHL} t_{PLH}	3003 (Fig. 5)	51 52	IN	OUT		5.0 v		IN	2.4 v	2.4 v	2.4 v	OUT	GND	2.4 v	2.4 v	2.4 v	1A to 1Y 2A to 2Y		1 14	
	t_{PLH} t_{PLH}		53 54	IN	OUT				IN	2.4 v	2.4 v	2.4 v	OUT		2.4 v	2.4 v	2.4 v	1A to 1Y 2A to 2Y		15 15	
11 T_A = -55°C	t_{PHL} t_{PHL}	3003 (Fig. 5)	55 56	IN	OUT		5.0 v		IN	2.4 v	2.4 v	2.4 v	OUT	GND	2.4 v	2.4 v	2.4 v	1A to 1Y 2A to 2Y		16 16	
	t_{PLH} t_{PLH}		57 58	IN	OUT				IN	2.4 v	2.4 v	2.4 v	OUT		2.4 v	2.4 v	2.4 v	1A to 1Y 2A to 2Y		13 13	

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TABLE III. Group A inspection for device type 08.
Terminal conditions (pins not designated are open)

Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 2 / 1B	3 / 12 / 1Y	4 / 14 / VCC	5 / 6 / 2Y	6 / 5 / 2A	7 / 4 / 2B	8 / 3 / 2C	9 / 9 / 3A	10 / 10 / 3B	11 / 7 / GND	12 / 11 / 3C	13 / 8 / 3Y	14 / 13 / 1C	Signal terminal	Test Limits Min	Max	Unit
V_{OL}	3007	1	2.0 v	2.0 v	20 mA	4.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND	5.5 v		2.0 v	1Y		0.4	V
V_{OL}		2	5.5 v	5.5 v		4.5 v	20 mA	2.0 v	2.0 v	2.0 v	5.5 v	5.5 v		5.5 v		5.5 v	2Y		0.4	
V_{OL}		3	5.5 v	5.5 v		4.5 v		5.5 v	5.5 v	5.5 v	2.0 v	2.0 v		2.0 v	20 mA		3Y		0.4	
V_{OH}	3006	4	0.8 v	5.5 v	-.5 mA	4.5 v					5.5 v	5.5 v		5.5 v			1Y	2.4		
V_{OH}		5	5.5 v	0.8 v	-.5 mA	4.5 v										0.8 v	1Y			
V_{OH}		6		5.5 v	-.5 mA	4.5 v										5.5 v	1Y			
V_{OH}		7					-.5 mA	0.8 v									2Y			
V_{OH}		8					-.5 mA	5.5 v	0.8 v								2Y			
V_{OH}		9					-.5 mA		5.5 v	0.8 v							2Y			
V_{OH}		10								5.5 v	0.8 v				-.5 mA		3Y			
V_{OH}		11									5.5 v	0.8 v			-.5 mA		3Y			
V_{OH}		12									5.5 v	5.5 v		0.8 v	-.5 mA		3Y			
I_{OS}	3011	13	GND	GND	GND	5.5 v										GND	1Y	-40	-100	mA
I_{OS}		14					GND	GND	GND	GND							2Y	-40	-100	mA
I_{OS}		15									GND	GND		GND	GND		3Y	-40	-100	mA
I_{IH1}	3010	16	2.4 v	GND				GND	GND	GND	GND	GND		GND		GND	1A		50	μA
I_{IH1}		17	GND	2.4 v												GND	1B			
I_{IH1}		18		GND												2.4 v	1C			
I_{IH1}		19						2.4 v								GND	2A			
I_{IH1}		20						GND	2.4 v								2B			
I_{IH1}		21							GND	2.4 v							2C			
I_{IH1}		22								GND	2.4 v						2A			
I_{IH1}		23									GND	2.4 v					2B			
I_{IH1}		24										GND		2.4 v			2C			
I_{IH2}		25	5.5 v	GND		5.5 v								GND			1A		100	
I_{IH2}		26	GND	5.5 v													1B			
I_{IH2}		27		GND				5.5 v								5.5 v	1C			
I_{IH2}		28						GND	5.5 v							GND	2A			
I_{IH2}		29							GND	5.5 v							2B			
I_{IH2}		30								GND	5.5 v						2C			
I_{IH2}		31									5.5 v	5.5 v					2A			
I_{IH2}		32									GND	GND					2B			
I_{IH2}		33									GND	GND		5.5 v			2C			
I_{IL}	3009	34	0.4 v	5.5 v		5.5 v		5.5 v	5.5 v	5.5 v	5.5 v	5.5 v	GND	5.5 v		5.5 v	1A	-1.0	-2.0	mA
		35	5.5 v	0.4 v												5.5 v	1B			
		36		5.5 v												0.4 v	1C			
		37						0.4 v								5.5 v	2A			
		38						5.5 v	0.4 v								2B			
		39							5.5 v	0.4 v							2C			
		40								5.5 v	0.4 v						2A			
		41									5.5 v	0.4 v					2B			
		42									5.5 v	5.5 v		5.4 v			2C			
I_{CCH}	3005	43	GND	GND				GND	GND	GND	GND	GND		GND		GND	VCC		12.6	
I_{CCL}	3006	44	5.5 v	5.5 v				5.5 v	5.5 v	5.5 v	5.5 v	5.5 v		5.5 v		5.5 v	VCC		30	

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 2 / 1B	3 / 12 / 1Y	4 / 14 / VCC	5 / 6 / 2Y	6 / 5 / 2A	7 / 4 / 2B	8 / 5 / 2C	9 / 9 / 3A	10 / 10 / 3B	11 / 7 / GND	12 / 11 / 3C	13 / 8 / 3Y	14 / 13 / 1C	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{IC}		45 46 47 48 49 50 51 52 53	-12 mA	-12 mA		4.5 v		-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	GND	-12 mA		-12 mA	1A 1B 1C 2A 2B 2C 3A 3B 3C		-1.5	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
9 T_A = 25°C	t_{PHL} t_{PHL} t_{PHL}	3003 (Fig. 5)	54 55 56	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	2.4 v	IN	2.4 v	GND	2.4 v	OUT	2.4 v	1A to 1Y 2A to 2Y 3A to 3Y	3	12 12 12	ns
	t_{PLH} t_{PLH} t_{PLH}		57 58 59	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y 2A to 2Y 3A to 3Y		12 12 12	
10 T_A = 125°C	t_{PHL} t_{PHL} t_{PHL}	3003 (Fig. 5)	60 61 62	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y 2A to 2Y 3A to 3Y		14 14 14	
	t_{PLH} t_{PLH} t_{PLH}		63 64 65	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y 2A to 2Y 3A to 3Y		15 15 15	
11 T_A = 55°C	t_{PHL} t_{PHL} t_{PHL}	3003 (Fig. 5)	66 67 68	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y 2A to 2Y 3A to 3Y		16 16 16	
7	t_{PLH} t_{PLH} t_{PLH}		69 70 71	IN	2.4 v	OUT		OUT	IN	2.4 v	2.4 v	IN	2.4 v		2.4 v	OUT	2.4 v	1A to 1Y 2A to 2Y 3A to 3Y		13 13 13	

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 2 / 1B	3 / 3 / 1Y	4 / 14 / VCC	5 / 6 / 2Y	6 / 4 / 2A	7 / 5 / 2B	8 / 6 / 3Y	9 / 9 / 3A	10 / 10 / 3B	11 / 7 / GND	12 / 12 / 4A	13 / 13 / 4B	14 / 11 / 4Y	Meas. terminal	Test limits: Min	Max	Unit
1 T_A = 25°C	V_{OL}	3007	1	2.0 v	2.0 v	20 mA	4.5 v	20 mA	5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		.4	V
	V_{OL}		2	5.5 v	5.5 v				2.0 v	2.0 v		6.5 v	5.5 v					2Y		.4	
	V_{OL}		3						5.5 v	5.5 v	20 mA	2.0 v	2.0 v					3Y		.4	
	V_{OL}		4									5.5 v	5.5 v		2.0 v	2.0 v	20 mA	4Y		4	
	V_{OH}	3006	5	.8 v	5.5 v	-.5 mA									5.5 v	5.5 v		1Y	2.4		
	V_{OH}		6	5.5 v	.8 v	-8 mA												1Y			
	V_{OH}		7	5.5 v	5.5 v			-.5 mA	.8 v	.8 v								2Y			
	V_{OH}		8					-8 mA	5.5 v	5.5 v	-.5 mA	.8 v	5.5 v					2Y			
	V_{OH}		9								-8 mA	5.5 v	.8 v					3Y			
	V_{OH}		10										5.5 v					3Y			
	V_{OH}		11												.8 v	5.5 v	-.5 mA	4Y			
	V_{OH}		12												5.5 v	.8 v	-.5 mA	4Y			
	I_{OS}	3011	13	GND	GND	GND	5.5 v	GND	GND	GND	GND	GND	GND					1Y	-40	-100	mA
	I_{OS}		14															2Y			
	I_{OS}		15															3Y			
	I_{OS}		16												GND	GND	GND	4Y			
	I_{IH1}	3010	17	2.4 v	GND				GND	GND		GND	GND		GND	GND		1A		50	μA
	I_{IH1}		18	GND	2.4 v				GND	GND								1B			
	I_{IH1}		19		GND				2.4 v	2.4 v								2A			
	I_{IH1}		20						GND	GND		2.4 v	2.4 v					2B			
	I_{IH1}		21									GND	GND					3A			
	I_{IH1}		22												2.4 v			3B			
	I_{IH1}		23												GND	2.4 v		4A			
	I_{IH1}		24															4B			
	I_{IH2}		25	5.5 v												GND		1A		100	
	I_{IH2}		26	GND	5.5 v				5.5 v	5.5 v								1B			
	I_{IH2}		27		GND				GND	GND		5.5 v	5.5 v					2A			
	I_{IH2}		28									GND	GND		5.5 v			2B			
	I_{IH2}		29												GND	5.5 v		3A			
	I_{IH2}		30															3B			
	I_{IH2}		31															4A			
	I_{IH2}		32															4B			
	I_{IL}	3009	33	0.4 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1A	-1.0	-2.0	mA
	I_{IL}		34	5.5 v	0.4 v				5.5 v	5.5 v								1B			
	I_{IL}		35		5.5 v				0.4 v	5.5 v								2A			
	I_{IL}		36						5.5 v	0.4 v		0.4 v	5.5 v					2B			
	I_{IL}		37							5.5 v		5.5 v	0.4 v					3A			
	I_{IL}		38										5.5 v		0.4 v	0.4 v		3B			
	I_{IL}		39															4A			
	I_{IL}		40												5.5 v			4B			

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	2	3	14	6	4	5	6	8	9	10	7	12	13	11		Min	Max	Unit
			Test No.	1A	1B	1Y	VCC	3Y	2A	2B	3Y	3A	3B	GND	4A	4B	4Y				
1 T_A = 25°C	I_{CCH}	3005	41	GND	GND		5.5 v		GND	GND		GND	GND	GND	GND	GND		VCC		10.8	mA
	I_{CCL}	3005	42	5.5 v	5.5 v		5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		VCC		40	mA
	V_{IC}		43 44 45 46 47 48 49 50	-12 mA	-12 mA		4.5 v		-12 mA	-12 mA		-12 mA	-12 mA		-12 mA	-12 mA		1A 1B 2A 2B 3A 3B 4A 4B		-1.5	V
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
9 T_A = 25°C	t_{PHL}	3003 (Fig. 5)	51 52 53 54	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y	2	12	ns
	t_{PLH}		55 56 57 58	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		12	
10 T_A = 125°C	t_{PHL}	3003 (Fig. 5)	59 60 61 62	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		14	
	t_{PLH}		63 64 65 66	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		15	
11 T_A = -55°C	t_{PHL}	3003 (Fig. 5)	67 68 69 70	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		18	
	t_{PLH}		71 72 73 74	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y		13	

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Meas. terminal	Test limits		
			Case C	1	4	3	14	5	6	9	8	11	10	7	12	13	2		Min	Max	Unit
			Test No.	1A	2Y	2A	VCC	3A	3Y	4A	4Y	5A	5Y	GND	6Y	6A	1Y				
1 T_A = 25°C	V_{OL}	3007	1	2.0 v		5.5 v	4.5 v	5.5 v		5.5 v		5.5 v		GND		5.5 v	20 mA	1Y		0.4	V
	V_{OL}		2	5.5 v	20 mA	2.0 v		5.5 v	20 mA									2Y			
	V_{OL}		3			5.5 v		2.0 v		2.0 v	20 mA							3Y			
	V_{OL}		4					5.5 v		5.5 v		2.0 v	20 mA					4Y			
	V_{OL}		5									5.5 v						5Y			
	V_{OL}		6												20 mA	2.0 v		6Y			
	V_{OH}	3006	7	0.5 v		5.5 v										5.5 v	-.5 mA	1Y	2.4		
	V_{OH}		8	5.5 v	-.5 mA	0.8 v		0.8 v	-.5 mA									2Y			
	V_{OH}		9			5.5 v		5.5 v		0.8 v	-.5 mA							3Y			
	V_{OH}		10							5.5 v		0.8 v	-.5 mA					4Y			
	V_{OH}		11									5.5 v			-.5 mA	0.8 v		5Y			
	V_{OH}		12															6Y			
	I_{OS}	3011	13	GND		GND	5.5 v										GND	1Y	-40	-100	mA
	I_{OS}		14		GND			GND	GND									2Y			
	I_{OS}		15							GND	GND							3Y			
	I_{OS}		16									GND	GND					4Y			
	I_{OS}		17												GND	GND		5Y			
	I_{OS}		18															6Y			
	I_{IH1}	3010	19	2.4 v		GND		GND		GND		GND						1A		50	μA
	I_{IH1}		20	GND		2.4 v		GND										2A			
	I_{IH1}		21			GND		2.4 v										3A			
	I_{IH1}		22					GND		2.4 v								4A			
	I_{IH1}		23							GND		2.4 v						5A			
	I_{IH1}		24									GND				2.4 v		6A			
	I_{IH2}		25	5.5 v		5.5 v		5.5 v								GND		1A		100	
	I_{IH2}		26	GND		GND		GND										2A			
	I_{IH2}		27					5.5 v		5.5 v								3A			
	I_{IH2}		28					GND		GND		5.5 v						4A			
	I_{IH2}		29							GND		GND						5A			
	I_{IH2}		30													5.5 v		6A			
	I_{IL}	3009	31	0.4 v		5.5 v		5.5 v		5.5 v		5.5 v						1A	-1.0	-3.0	mA
	I_{IL}		32	5.5 v		0.4 v		5.5 v										2A			
	I_{IL}		33			5.5 v		0.4 v		0.4 v		0.4 v						3A			
	I_{IL}		34					5.5 v		5.5 v								4A			
	I_{IL}		35							0.4 v		0.4 v						5A			
	I_{IL}		36							5.5 v		5.5 v				0.4 v		6A			
	I_{CCL}	3005	37	5.5 v		5.5 v	5.5 v	5.5 v		5.5 v		5.5 v		GND		5.5 v		VCC		60	mA
	I_{CCH}	3005	38	GND		GND	5.5 v	GND		GND		GND				GND		VCC		36	mA
	V_{IC}		39	-12 mA			4.5 v											1A		-1.5	V
			40			-12 mA												2A			
			41					-12 mA										3A			
			42							-12 mA								4A			
			43									-12 mA						5A			
			44													-12 mA		6A			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 4 / 1Y	3 / 3 / 2A	4 / 14 / VCC	5 / 5 / 2A	6 / 6 / 2Y	7 / 9 / 3A	8 / 8 / 3Y	9 / 11 / 4A	10 / 10 / 4Y	11 / 7 / GND	12 / 12 / 5Y	13 / 13 / 5A	14 / 2 / 1Y	Meas. terminal	Test limits Min	Max	Unit
9 T_A = 25°C	t_{PHL}	3003 (Fig. 5)	45 46 47 48 49 50	IN	OUT	IN	5.0 v	IN	OUT	IN	OUT	IN	OUT	GND	OUT	IN	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y	2	12	ns
	t_{PLH}		51 52 53 54 55 56	IN	OUT	IN		IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y		12	
10 T_A = 125°C	t_{PHL} t_{PHL} t_{PHL} t_{PHL} t_{PHL} t_{PHL}	3003 (Fig. 5)	57 58 59 60 61 62	IN	OUT	IN	5.0 v	IN	OUT	IN	OUT	IN	OUT	GND	OUT	IN	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y		14	
	t_{PLH} t_{PLH} t_{PLH} t_{PLH} t_{PLH} t_{PLH}		63 64 65 66 67 68	IN	OUT	IN		IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y		15	
11 T_A = -55°C	t_{PHL} t_{PHL} t_{PHL} t_{PHL} t_{PHL} t_{PHL}	3003 (Fig. 5)	69 70 71 72 73 74	IN	OUT	IN	5.0 v	IN	OUT	IN	OUT	IN	OUT	GND	OUT	IN	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y		16	
8	t_{PLH} t_{PLH} t_{PLH} t_{PLH} t_{PLH} t_{PLH}		75 76 77 78 79 80	IN	OUT	IN		IN	OUT	IN	OUT	IN	OUT		OUT	IN	OUT	1A to 1Y 2A to 2Y 3A to 3Y 4A to 4Y 5A to 5Y 6A to 6Y		13	

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 / 1 / 1A	2 / 2 / 1B	3 / 3 / 1Y	4 / 14 / VCC	5 / 6 / 2Y	6 / 4 / 2A	7 / 5 / 2B	8 / 8 / 3Y	9 / 9 / 3A	10 / 10 / 3B	11 / 7 / GND	12 / 12 / 4A	13 / 13 / 4B	14 / 11 / 4Y	Meas terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OL}	3007	1	2.0 v	2.0 v	20 mA	4.5 v		5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		1Y		0.4	V
			2	5.5 v	5.5 v			20 mA	2.0 v	2.0 v		5.5 v	5.5 v					2Y		0.4	
			3						5.5 v	5.5 v	20 mA	2.0 v	2.0 v					3Y		0.4	
			4									5.5 v	5.5 v		2.0 v	2.0 v	20 mA	4Y		0.4	
	I_{CEX}		5	0.8 v	4.5 v	5.5 v						5.5 v	5.5 v		5.5 v	5.5 v		1Y		250	µA
			6	4.5 v	0.8 v	5.5 v												1Y		250	
			7	5.5 v	5.5 v			5.5 v	0.8 v	4.5 v								2Y		250	
			8					5.5 v	4.5 v	0.8 v								2Y		250	
			9						5.5 v	5.5 v	5.5 v	0.8 v	4.5 v					3Y		250	
			10								5.5 v	4.5 v	0.8 v					3Y		250	
			11												0.8 v	4.5 v	5.5 v	4Y		250	
			12												4.5 v	0.8 v	5.5 v	4Y		250	
	V_{IC}		13	-12 mA														1A	-1.5		V
			14		-12 mA													1B	-1.5		
			15						-12 mA									2A	-1.5		
			16							-12 mA								2B	-1.5		
			17									-12 mA						3A	-1.5		
			18										-12 mA					3B	-1.5		
			19												-12 mA			4A	-1.5		
			20													-12 mA		4B	-1.5		
	I_{IH1}	3010	21	2.4 v	GND		5.5 v		GND	GND		GND	GND		GND	GND		1A		50	µA
			22	GND	2.4 v				GND									1B		50	
			23		GND				2.4 v									2A		50	
			24						GND	2.4 v								2B		50	
			25							GND		2.4 v						3A		50	
			26									GND	2.4 v					3B		50	
			27										GND		2.4 v			4A		50	
			28												GND	2.4 v		4B		50	
	I_{IH2}		29	5.5 v												GND		1A		100	
			30	GND	5.5 v													1B		100	
			31		GND				5.5 v									2A		100	
			32						GND	5.5 v								2B		100	
			33							GND		5.5 v						3A		100	
			34									GND	5.5 v					3B		100	
			35										GND		5.5 v			4A		100	
			36												GND	5.5 v		4B		100	
	I_{IL}	3009	37	0.4 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v		5.5 v	5.5 v		1A	-1.0	-2.0	mA
			38	5.5 v	0.4 v				5.5 v									1B	-1.0	-2.0	
			39		5.5 v				0.4 v									2A	-1.0	-2.0	
			40						5.5 v	0.4 v								2B	-1.0	-2.0	
			41							5.5 v		0.4 v	5.5 v					3A	-1.0	-2.0	
			42									5.5 v	0.4 v					3B	-1.0	-2.0	
			43										5.5 v		0.4 v			4A	-1.0	-2.0	
			44												5.5 v	0.4 v		4B	-1.0	-2.0	
	I_{CCH}	3005	45	GND	GND				GND	GND		GND	GND	GND	GND	GND		VCC		16.8	mA
	I_{CCL}	3005	46	5.5 v	5.5 v				5.5 v	5.5 v		5.5 v	5.5 v	GND	5.5 v	5.5 v		VCC		40	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / 1A	2 / 2 / 1B	3 / 3 / 1Y	4 / 14 / VCC	5 / 8 / 2Y	6 / 4 / 2A	7 / 5 / 2B	8 / 6 / 3Y	9 / 9 / 3A	10 / 10 / 3B	11 / 7 / GND	12 / 12 / 4A	13 / 13 / 4B	14 / 11 / 4Y	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
9 T_A = 25°C	t_{PHL}	3003 (Fig. 6)	47	IN	2.4 v	OUT	5.0 v	OUT	IN	2.4 v	OUT	IN	2.4 v	GND	IN	2.4 v	OUT	1A to 1Y	3	14	ns
			48															2A to 2Y	3	14	
			49															3A to 3Y	3	14	
			50															4A to 4Y	3	14	
	t_{PLH}		51	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y	3	17	
			52															2A to 2Y	3	17	
			53															3A to 3Y	3	17	
			54															4A to 4Y	3	17	
10 T_A = 125°C	t_{PHL}	3003 (Fig. 6)	55	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y	3	16	
			56															2A to 2Y	3	16	
			57															3A to 3Y	3	16	
			58															4A to 4Y	3	16	
	t_{PLH}		59	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y	3	21	
			60															2A to 2Y	3	21	
			61															3A to 3Y	3	21	
			62															4A to 4Y	3	21	
11 T_A = -55°C	t_{PHL}	3003 (Fig. 6)	63	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y	3	18	
			64															2A to 2Y	3	18	
			65															3A to 3Y	3	18	
			66															4A to 4Y	3	18	
	t_{PLH}		67	IN	2.4 v	OUT		OUT	IN	2.4 v	OUT	IN	2.4 v		IN	2.4 v	OUT	1A to 1Y	3	19	
			68															2A to 2Y	3	19	
			69															3A to 3Y	3	19	
			70															4A to 4Y	3	19	

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TABLE III. Group A inspection for device type 07.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Unit A, E, A Case C Test No	1 1 LA	2 3 1Y	3 5 NC	4 14 VCC	5 11 NC	6 9 2A	7 10 2B	8 12 2C	9 13 2D	10 8 2Y	11 7 GND	12 2 1B	13 4 1C	14 5 1D	Meas terminal	Test limits Min	Max	Unit
1	V_{OL}	3007	1	2.0 v	30 mA		4.5 v		5.5 v	5.5 v	5.5 v	5.5 v		GND	2.0 v	2.0 v	2.0 v	1Y		0.4	V
T_A = 25°C	V_{OL}	3007	2	5.5 v					2.0 v	2.0 v	2.0 v	2.0 v	20 mA		5.5 v	5.5 v	5.5 v	2Y		0.4	V
	I_{CEX}		3	0.8 v	5.5 v													1Y		250	µA
			4	5.5 v											0.8 v	5.5 v	5.5 v	1Y			
			5	5.5 v											5.5 v	0.8 v	5.5 v	1Y			
			6															1Y			
			7						0.8 v				5.5 v					2Y			
			8						5.5 v	0.8 v								2Y			
			9							5.5 v	0.8 v							2Y			
			10								5.5 v	0.8 v						2Y			
	V_{IC}		11	-12 mA														1A		-1.5	V
			12												-12 mA			1B			
			13													-12 mA		1C			
			14														-12 mA	1D			
			15						-2 mA									2A			
			16							-12 mA								2B			
			17								-12 mA							2C			
			18									-12 mA						2D			
	I_{IH1}	3010	19	2.4 v					GND	GND	GND	GND			GND	GND	GND	1A		50	µA
			20	GND											2.4 v	GND		1B			
			21												GND	2.4 v	2.4 v	1C			
			22													GND		1D			
			23														2.4 v	2A			
			24						2.4 v	2.4 v	2.4 v	2.4 v					GND	2B			
			25						GND	GND	GND	GND						2C			
			26															2D			
	I_{IH2}		27	5.5 v					GND	GND	GND	GND			GND	GND	GND	1A		100	µA
			28	GND											5.5 v	GND		1B			
			29												GND	5.5 v	5.5 v	1C			
			30													GND		1D			
			31						5.5 v	5.5 v	5.5 v						5.5 v	2A			
			32						GND	GND	GND	5.5 v					GND	2B			
			33															2C			
			34															2D			
	I_{IL}	3009	35	0.4 v					5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	5.5 v	5.5 v	1A	1.0	2.0	mA
			36	5.5 v											0.4 v	5.5 v		1B			
			37	5.5 v											5.5 v	0.4 v	5.5 v	1C			
			38	5.5 v			5.5 v		5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	5.5 v	0.4 v	1D	1.0	2.0	mA
			39						0.4 v	0.4 v							5.5 v	2A			
			40						5.5 v	5.5 v	0.4 v	0.4 v						2B			
			41								5.5 v	5.5 v						2C			
			42															2D			
	I_{CCL}	3005	43	5.5 v					5.5 v	5.5 v	5.5 v	5.5 v			5.5 v	5.5 v	5.5 v	VCC		20	mA
	I_{CCH}	3005	44	GND					GND	GND	GND	GND			GND	GND	GND	VCC		8.4	mA

2 Same tests, terminal conditions and limits as for subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.

3 Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated are open)

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 / 1 LA	2 / 6 1Y	3 / 3 NC	4 / 14 VCC	5 / 11 NC	6 / 9 2A	7 / 10 2B	8 / 12 2C	9 / 13 2D	10 / 8 2Y	11 / 7 GND	12 / 2 1B	13 / 4 1C	14 / 5 1D	Meas terminal	Test limits Min	Max	Unit
9 T_A = 25°C	t_{PHL}	3003 (Fig. 6)	45	IN	OUT		5.0 v							GND				1A to 1Y	3	14	ns
	t_{PHL}		46						IN	2.4 v	2.4 v	2.4 v	OUT		2.4 v	2.4 v	2.4 v	2A to 1Y		14	
	t_{PLH}		47	IN	OUT													1A to 1Y		17	
	t_{PLH}		48						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		17	
10 T_A = 125°C	t_{PHL}	3003 (Fig. 6)	49	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		16	
	t_{PHL}		50						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		16	
	t_{PLH}		51	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		21	
	t_{PLH}		52						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		21	
11 T_A = -55°C	t_{PHL}	3003 (Fig. 6)	53	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		18	
	t_{PHL}		54						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		18	
	t_{PLH}		55	IN	OUT										2.4 v	2.4 v	2.4 v	1A to 1Y		19	
	t_{PLH}		56						IN	2.4 v	2.4 v	2.4 v	OUT					2A to 2Y		19	

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4.4.3 Group C inspection. Group C inspection shall consist of the test subgroups and LTPD values shown in table III of Method 5005 of MIL-STD-883 and as follows:

(a) End point electrical parameters shall be as specified in table II.
(b) Subgroups 7 and 8 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
(c) Lead bend in only one direction is required for initial conditioning prior to moisture resistance and salt atmosphere tests.
(d) High temperature storage test (Method 1008 of MIL-STD-883) conditions:
  (1) Temperature: 150 ±10°C.
  (2) Duration: 1,000 hours, except as otherwise permitted by Appendix B of MIL-M-38510.
(e) Operating life test (Method 1005 of MIL-STD-883) conditions:
  (1) Test condition D or E, using the circuit shown on figure 4, or equivalent.
  (2) T_A = 125°C, minimum.
  (3) Test duration: 1,000 hours, except as permitted by Appendix B of MIL-M-38510.
(f) Omit steady state reverse bias test.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of preparation for delivery. Inspection of preparation for delivery shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging and packing. Microcircuits shall be prepared for delivery in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. The contract or order should specify the following.

(a) Complete part number (see 1.2).
(b) Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
(c) Requirement for certificate of compliance, if applicable.
(d) Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
(e) Requirements for packaging and packing.
(f) Requirements for failure analysis (including required test condition of Method 5003), corrective action and reporting of results, if applicable.

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(g) Requirements for product assurance options.
(h) Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - - - - - - - Voltage level at an input terminal
V_{IC} - - - - - - - - - - - - Input clamp voltage
I_{IN} - - - - - - - - - - - - Current flowing into an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3), and case outline A (see 1.2.3). There is no substitute for case outline C. Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. Microcircuits covered by this specification are substitutable for the following commercial device types:

Device type	Commercial type
01	54H30
02	54H20
03	54H10
04	54H00
05	54H04
06	54H01
07	54H22

Custodians:
Air Force - 17
Army - EL
Navy - EC
NASA - NA

Review activities:
Army - EL, MI
Navy -
Air Force - 11, 70, 80
DSA - ES

User activities:
Army - SM, MU, WC
Navy - AS, OS, CG, MC
Air Force - 13, 15, 19

Preparing activity:
Air Force - 17

Agent:
DSA - ES

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